

Foundation (Jalapeno)

- Q1.** Simplify $2^3 \cdot 2^4$
(A) 2^{3+4}
(B) $2^{3 \cdot 4}$
(C) 2^{3-4}
(D) $2^{3/4}$
(E) $2^{4/3}$
- Q2.** A basketball player scores 3^2 points in the first quarter and 3^3 points in the second quarter. What is the total number of points scored?
(A) 3^4
(B) 3^5
(C) 3^{2+3}
(D) $3^{2 \cdot 3}$
(E) 3^{3-2}
- Q3.** Simplify $(4^2)^3$
(A) $4^{2 \cdot 3}$
(B) 4^{2+3}
(C) 4^{2-3}
(D) $4^{3/2}$
(E) $4^{2/3}$
- Q4.** A cyclist travels 2^4 miles in the morning and 2^5 miles in the afternoon. What is the total distance traveled?
(A) 2^{4+5}
(B) $2^{4 \cdot 5}$
(C) 2^{5-4}
(D) $2^{4/5}$
(E) $2^{5/4}$
- Q5.** Simplify $3^4 \div 3^2$
(A) 3^{4-2}
(B) 3^{4+2}
(C) $3^{4 \cdot 2}$
(D) $3^{4/2}$
(E) $3^{2/4}$
- Q6.** A tennis player hits 5^2 aces in a match and 5^3 aces in the next match. What is the total number of aces hit?
(A) 5^{2+3}
(B) $5^{2 \cdot 3}$
(C) 5^{3-2}
(D) $5^{2/3}$
(E) $5^{3/2}$
- Q7.** Simplify $(2^3)^2$
(A) $2^{3 \cdot 2}$
(B) 2^{3+2}
(C) 2^{3-2}
(D) $2^{2/3}$
(E) $2^{3/2}$
- Q8.** A swimmer completes 4^3 laps in the morning and 4^4 laps in the afternoon. What is the total number of laps completed?
(A) 4^{3+4}
(B) $4^{3 \cdot 4}$
(C) 4^{4-3}
(D) $4^{3/4}$
(E) $4^{4/3}$

Open Ended Questions (FRQ)

- Q9.** Simplify the expression $2^3 \cdot 2^4 \div 2^2$ and show your work.

- Q10.** A golfer scores 3^2 points on the first hole, 3^3 points on the second hole, and 3^4 points on the third hole. What is the total number of points scored? Show your work.

Chef's Tips

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- Tip 1: Ensure students understand the rules of exponents before attempting the test.
- Tip 2: Encourage students to show their work for free-response questions.
- Tip 3: Provide feedback on student answers to help them improve their understanding of exponents.